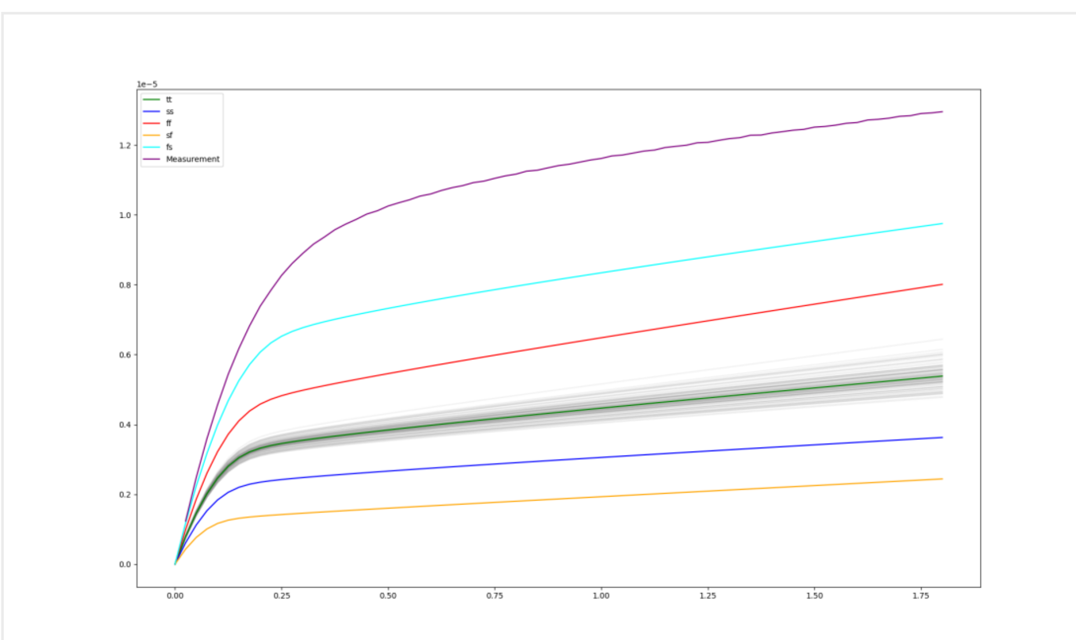
Results and discussion from Discord’s “521 Bias Generator - Tiny Tapeout” thread:

rburt16 reported @Rod Burt's Bias Generator as working: It works! The measurements of the 16 current sources have an i_out (mean) = 0.7uA and an i_out (std) = 0.1uA. These results also match measurements taken on tt08-silicon by smunaut. Current source layout orientation shifted by 90 degrees does not show a statistical difference. The difference between the target i_out = 1uA and the actual 0.7uA is unexpected. Although, that is the reason to run a test chip. This could be due to modeling issues for devices running near sub-threshold or possibly a layout issue. More investigation is needed. The wide spread in current actually matches monti-carlo simulations reasonably well. At the time of tape out I had not run monti-carlo simulations yet, but have since run them. I have attached simulation results that also include recommendations for improvements. Basically, the length and corresponding width of key devices need to be significantly increased to reduce the chip to chip variation. The tradeoff will be increased cell size.

tnt [TT],
BTW, I used a project called "Raw Transistor" which exposes just .. well a few raw transistor (G/D/S) directly to plot curves and compare to the models.
Results are that definitely modelling isn't great when you're anywhere close to threshold. And results are more like the fs corner than tt. (Also contrary to convention fs in sky130 is fast pmos, slow nmos ...)

Rod Burt
Thanks tnt! The fs corner simulations move I_out to 0.91uA. This is part of the problem. I’ll take a look at your “Raw Transistor” project results for other potential insights.

tnt [TT],
See this for instance.



This is a Vds/Ids sweep of a 1v8 pmos svt at Vgs=-1V
You can see the measurement is way off from the model. fs is the closest but ... not quite there.

meas_tt08_2026023.tar.bz2
112.28 KB

This is the raw measurement results.

Rod Burt
I see what you mean. Thanks for the raw data.